IGCSE 0607 Practice Questions — E4.4 — Combined Questions & Markscheme

Equations of linear graphs

Total Questions: 15 — **Total Marks:** 102 Marks

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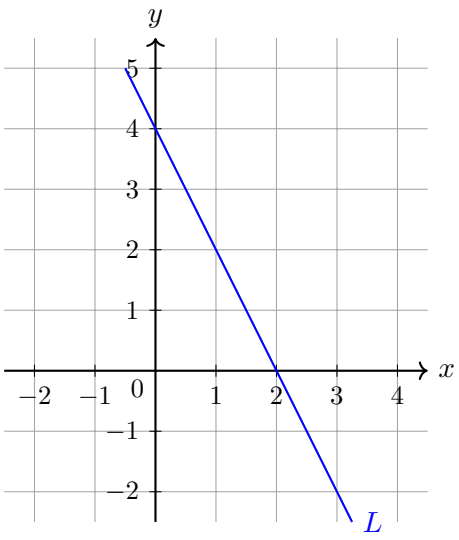
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1 Find the gradient and the y-intercept of the line with equation

$3x + 5y = 20$

Gradient = ... , y-intercept = ... **[3]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Write the equation of line L in the form $ax + by = c$, where a, b and c are integers.

..... **[1]**

3 **(a)** Find the gradient and the y-intercept of the line $3x + 5y = 15$.

Gradient = ... , y-intercept = ... **[3]**

(b) Find the equation of the line parallel to $3x + 5y = 15$ that passes through the point $(0, -4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

4 A straight line L has the equation $3x - 4y = 12$.

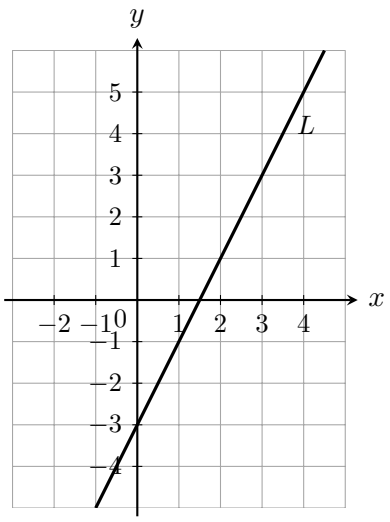
(a) Find the gradient and the y-intercept of line L.

Gradient = ... , y-intercept = ... **[3]**

(b) Another line M is parallel to L and passes through the point $(0, 5)$.
Write down the equation of line M in the form $y = mx + c$.

$y =$ ..... **[2]**

5 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the equation of line L in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) A vertical line M passes through the point $(-2, 4)$.
Write down the equation of line M.

..... **[1]**

(c) Find the coordinates of the point where line L and line M intersect.

(..... , ...) **[2]**

6 Line L_1 has equation $4x + 5y = 20$.

(a) Find the gradient and the y-intercept of line L_1.

Gradient = ... , y-intercept = ... **[3]**

(b) Find the coordinates of the point where line L_1 crosses the x-axis.

(..... , ...) **[2]**

(c) Line L_2 passes through the points $(-3, -5)$ and $(3, 7)$.
Find the equation of line L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

7 A is the point $(p, -1)$ and B is the point $(5, 2p - 1)$.
The gradient of the line AB is 3.

(a) Find the value of p.

$p =$ ..... **[3]**

(b) Find the equation of the line AB.
Give your answer in the form $y = mx + c$.

..... **[2]**

(c) The line AB intersects the x-axis at the point C and the y-axis at the point D.
Work out the area of triangle COD, where O is the origin.

..... **[3]**

8 A straight line L passes through the points $A(-2, 3k)$ and $B(4, k)$, where k is a non-zero constant.

(a) Find the gradient of line L in terms of k.
Give your answer in its simplest form.

..... **[2]**

(b) Find the equation of line L in the form $ax + by = c$, where a, b, and c are expressions in terms of k.

..... **[3]**

(c) Line L also passes through the point $(1, 10)$.
Find the value of k.

$k =$ ..... **[2]**

(d) Hence, write down the coordinates of the x-intercept of line L.

(..... , ...) **[2]**

9 The line L passes through the points $A(2a, a+4)$ and $B(a-1, 3a+2)$.
The gradient of line L is -4.

(a) Find the value of a.

$a =$ ..... **[3]**

(b) Find the equation of line L.
Give your answer in the form $px + qy = r$, where p, q and r are integers and $p > 0$.

..... **[3]**

(c) Line L intersects the line $y = 3x - 2$ at the point M.
Find the coordinates of M.

(..... , ...) **[3]**

10 A is the point $(-3, 1)$ and B is the point $(5, 5)$.

(a) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

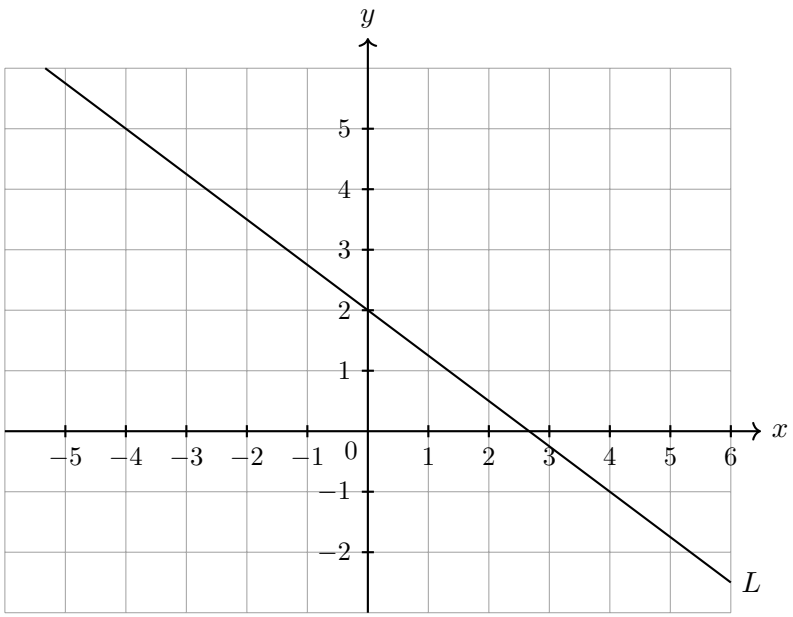
(b) The perpendicular bisector of AB intersects the line $x - 2y = 5$ at the point P.
Find the coordinates of P.

(..... , ...) **[3]**

(c) Find the exact length of AP.

..... **[3]**

11 The diagram shows a line L drawn on a coordinate grid.



(a) Work out the gradient of line L.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

12 Point A has coordinates $(-4, 3)$ and point B has coordinates $(6, -2)$.

(a) Find the equation of the line passing through A and B.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

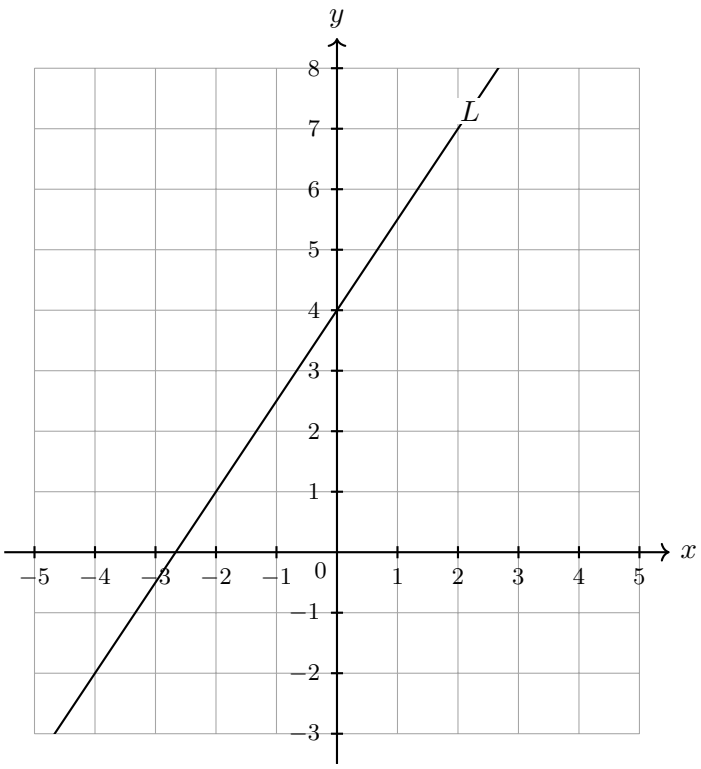
(b) The line $3x - 5y = 15$ intersects the x-axis at point C and the y-axis at point D.
Find the coordinates of C and D.

$C =$ ..... , $D =$ ..... **[2]**

(c) Write down the equation of the line that is parallel to the y-axis and passes through point A.

..... **[1]**

13 The diagram shows a straight line L.



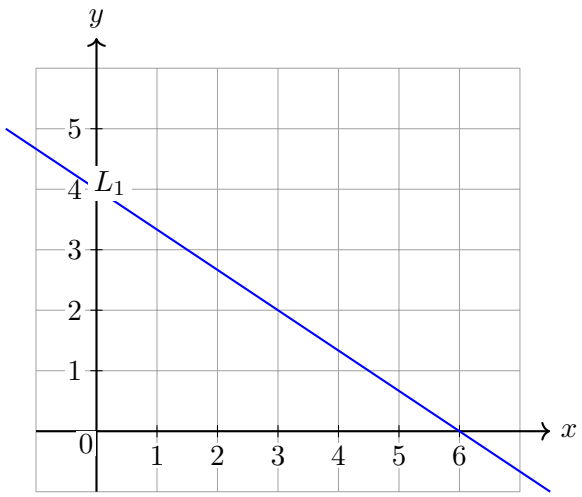
(a) Find the equation of line L.
Give your answer in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

(b) Line M is perpendicular to line L and passes through the point $(9, -1)$.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

14 The diagram shows a straight line L_1 drawn on a coordinate grid.



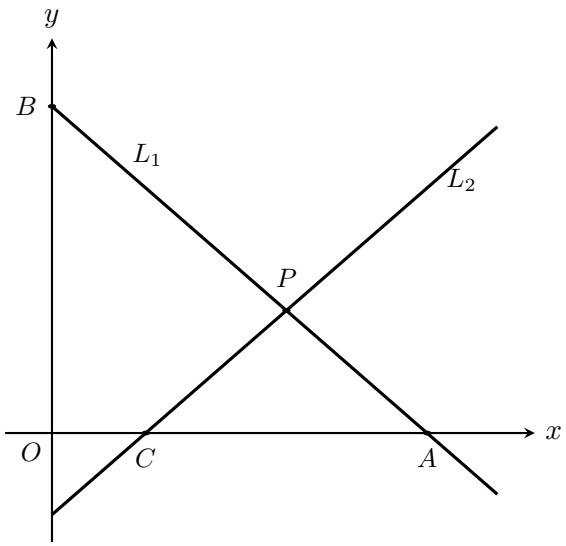
(a) Find the equation of line L_1 in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Line L_2 passes through the point $(2, 7)$ and the midpoint of the segment of L_1 that lies between the axes.
Find the equation of L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[4]**

15 The diagram shows a sketch of two straight lines, L_1 and L_2, which intersect at the point P.
Line L_1 has equation $3x + 2y = 24$ and crosses the x-axis at A and the y-axis at B.
Line L_2 has equation $y = 1.5x - 3$ and crosses the x-axis at C.



(a) Write down the coordinates of A, B, and C.

$A =$ ..... , $B =$ ..... , $C =$ ..... **[3]**

(b) Find the coordinates of the point of intersection P.

$P =$ (..... , ...) **[3]**

(c) Calculate the area of quadrilateral $OCPB$.

..... square units **[3]**

(d) A third line L_3 has equation $kx - 4y = 14$ and also passes through point P.
Calculate the value of k.

$k =$ ..... **[2]**

1 Rearranging to $5y = -3x + 20$ or $y = -\frac{3}{5}x + 4$ **[M1]**

Gradient $= -\frac{3}{5}$ oe **[A1]**

y-intercept $= 4$ **[A1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** Gradient $= -2$ **[B1]**

y-intercept $= 4$ **[B1]**

$-2x + 4$ **[B1]**

(b) $2x + y = 4$ oe **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Rearranges equation to $y = mx + c$ form: $5y = -3x + 15$ or $y = -\frac{3}{5}x + 3$ **[M1]**

Gradient $= -\frac{3}{5}$ **[A1]**

y-intercept $= 3$ **[A1]**

(b) Identifies parallel gradient as $-\frac{3}{5}$ **[M1]**

$-\frac{3}{5}x - 4$ **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** Rearranges the equation to make y the subject: $4y = 3x - 12$ or $y = \frac{3}{4}x - 3$ **[M1]**

Gradient $= \frac{3}{4}$ **[A1]**

y-intercept $= -3$ **[A1]**

(b) Identifies the gradient of M is $\frac{3}{4}$ **[M1]**

$\frac{3}{4}x + 5$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** Gradient $= \frac{3 - (-3)}{3 - 0} = 2$ **[M1]**

y-intercept $= -3$ **[B1]**

$2x - 3$ **[A1]**

(b) $x = -2$ **[B1]**

(c) $y = 2(-2) - 3 = -7$ **[M1]**

$(-2, -7)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** $5y = -4x + 20$ or $y = -\frac{4}{5}x + 4$ **[M1]**

Gradient $= -\frac{4}{5}$ (or -0.8) **[A1]**

y-intercept $= 4$ **[A1]**

(b) Substitute $y = 0$: $4x = 20$ **[M1]**

$(5, 0)$ **[A1]**

(c) Gradient $= \frac{7 - (-5)}{3 - (-3)} = 2$ **[M1]**

$y = 2x + 1$ **[M1]**

$2x - y = -1$ (or $-2x + y = 1$) **[A1]**

[Back to Question] *[Total Marks: 8]*

7 **(a)** $\frac{(2p - 1) - (-1)}{5 - p} = 3$ **[M1]**

$2p = 3(5 - p)$ **[M1]**

$p = 3$ **[A1]**

(b) Substituting $p = 3$ to find coordinates $A(3, -1)$ or $B(5, 5)$ and using gradient 3 to find equation **[M1]**

$y = 3x - 10$ **[A1]**

(c) Identifies x-intercept as $\frac{10}{3}$ or y-intercept as -10 **[M1]**

$\frac{1}{2} \times \frac{10}{3} \times 10$ **[M1]**

$\frac{50}{3}$ or 16.7 $(16.66\ldots)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

8 **(a)** Sets up the gradient expression: $\frac{k - 3k}{4 - (-2)}$ **[M1]**

$-\frac{k}{3}$ **[A1]**

(b) Substitutes a point and gradient into the linear equation: $y - k = -\frac{k}{3}(x - 4)$ or $y - 3k = -\frac{k}{3}(x - (-2))$ **[M1]**

Multiplies by 3 to eliminate the fraction: $3y - 3k = -kx + 4k$ **[M1]**

$kx + 3y = 7k$ **[A1]**

(c) Substitutes $x = 1$ and $y = 10$ into their equation from part (b): $k(1) + 3(10) = 7k$ **[M1]**

5 **[A1]**

(d) Substitutes $k = 5$ to find the line equation $5x + 3y = 35$ and sets $y = 0$: $5x = 35$ **[M1]**

$(7, 0)$ **[A1]**

[Back to Question] *[Total Marks: 9]*

9 (a) $\dfrac{(3a + 2) - (a + 4)}{(a - 1) - 2a} = -4$ **[M1]**

$\dfrac{2a - 2}{-a - 1} = -4$ **[M1]**

$2a - 2 = 4a + 4$

-6 = 2a

a = -3[A1]

(b) Identifies coordinates of A as $(-6, 1)$ or B as $(-4, -7)$ **[M1]**

$y - 1 = -4(x + 6)$ or $y = -4x - 23$ **[M1]**

$4x + y = -23$ **[A1]**

(c) $4x + (3x - 2) = -23$ or $-4x - 23 = 3x - 2$ oe **[M1]**

$7x = -21$

x = -3[A1]

$y = -11$ **[A1]**

[Back to Question] *[Total Marks: 9]*

10 (a) Calculates the midpoint of $AB = \left(\dfrac{-3 + 5}{2}, \dfrac{1 + 5}{2}\right) = (1, 3)$ **[B1]**

Calculates the gradient of $AB = \dfrac{5 - 1}{5 - (-3)} = \dfrac{4}{8} = \dfrac{1}{2}$ **[B1]**

Finds the perpendicular gradient $= -2$ **[M1]**

Uses perpendicular gradient -2 and midpoint $(1, 3)$ to find equation, e.g., $3 = -2(1) + c$ **[M1]**

$-2x + 5$ **[A1]**

(b) Sets up simultaneous equations, e.g., substituting $y = -2x + 5$ into $x - 2y = 5$ **[M1]**

$x - 2(-2x + 5) = 5 \implies 5x - 10 = 5 \implies x = 3$ **[M1]**

$(3, -1)$ **[A1]**

(c) Uses distance formula for $A(-3, 1)$ and $P(3, -1)$ **[M1]**

$\sqrt{(3 - (-3))^2 + (-1 - 1)^2}$ **[M1]**

$\sqrt{40}$ or $2\sqrt{10}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

11 **(a)** Evidence of using two integer coordinates on the line, e.g. $\frac{5-(-1)}{-4-4}$ **[M1]**

$-\frac{3}{4}$ or -0.75 **[A1]**

(b) $y = -\frac{3}{4}x + 2$ or $y - 5 = -\frac{3}{4}(x+4)$ **[M1]**

$4y = -3x + 8$ or $4y - 20 = -3x - 12$ **[M1]**

$3x + 4y = 8$ or equivalent integer form **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** Gradient $= \frac{-2-3}{6-(-4)}$ or $\frac{-5}{10} = -\frac{1}{2}$ **[M1]**

Substitution of $(-4, 3)$ or $(6, -2)$ into $y = -\frac{1}{2}x + c$, e.g. $3 = -\frac{1}{2}(-4) + c$ **[M1]**

$-\frac{1}{2}x + 1$ **[A1]**

(b) $C = (5, 0)$ **[B1]**

$D = (0, -3)$ **[B1]**

(c) $x = -4$ **[B1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** Identifies two correct points on the line, e.g., $(0, 4)$ and $(2, 7)$ **[M1]**

Calculates the gradient $m = \frac{7-4}{2-0} = \frac{3}{2}$ **[M1]**

States correct equation in required integer form, e.g., $3x - 2y = -8$ or $2y - 3x = 8$ **[A1]**

(b) Uses $m_1 \times m_2 = -1$ to find gradient of M: $-\frac{2}{3}$ **[M1]**

Substitutes $(9, -1)$ into $y = -\frac{2}{3}x + c$ to find c: $-1 = -\frac{2}{3}(9) + c$ **[M1]**

$-\frac{2}{3}x + 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** Identifies the y-intercept is 4 or uses coordinates $(0, 4)$ **[B1]**

Calculates the gradient: $\frac{0-4}{6-0}$ or $\frac{-4}{6}$ **[M1]**

$-\frac{2}{3}x + 4$ **[A1]**

(b) Finds the coordinates of the midpoint of the segment $(0, 4)$ to $(6, 0)$: $\left(\frac{0+6}{2}, \frac{4+0}{2}\right)$ or $(3, 2)$ **[B1]**

Calculates the gradient of L_2: $\frac{7-2}{2-3}$ or $\frac{5}{-1} = -5$ **[M1]**

Uses point-gradient form for L_2: $y - 7 = -5(x - 2)$ or $y - 2 = -5(x - 3)$ **[M1]**

$5x + y = 17$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $A = (8, 0)$ **[B1]**

$B = (0, 12)$ **[B1]**

$C = (2, 0)$ **[B1]**

(b) $3x + 2(1.5x - 3) = 24$ **[M1]**

$6x = 30$ giving $x = 5$ **[A1]**

$(5, 4.5)$ **[A1]**

(c) Area of $\triangle OAB = \frac{1}{2} \times 8 \times 12 = 48$ **[M1]**

Area of $\triangle CAP = \frac{1}{2} \times (8 - 2) \times 4.5 = 13.5$ **[M1]**

34.5 **[A1]**

(d) $5k - 4(4.5) = 14$ **[M1]**

6.4 **[A1]**

[Back to Question] *[Total Marks: 11]*